COMMUNITY HERO: HYPERLOCAL CIVIC PROBLEM SOLVER & SMART CITY INCIDENT RESOLUTION PLATFORM WITH 3D GEOSPATIAL VISUALIZATION AND MACHINE LEARNING

PROJECT REPORT [Phase-III]

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

Department of Computer Science and Engineering

Submitted to

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL (M.P.)



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CERTIFICATE

This is to certify that the work embodied in this report entitled "Community Hero: Hyperlocal Civic Problem Solver & Smart City Incident Resolution Platform with 3D Geospatial Visualization and Machine Learning" has been satisfactorily completed by AARUSH BHARTI (24U020010) & ASHISH GOUTAM (24U020011). It is a bonafide piece of work, carried out under our guidance in the Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhopal for the partial fulfillment of the Bachelor of Technology during the academic year 2026-27.

The web application, trained classification model, 75-meter spatial duplicate detection algorithm, voice intake system, and testing phases have been evaluated as part of the Phase III minor project deliverables.

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DECLARATION

We hereby declare that the following project report entitled “Community Hero: Hyperlocal Civic Problem Solver & Smart City Incident Resolution Platform with 3D Geospatial Visualization and Machine Learning” presented in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is an authentic documentation of our original work, carried out under the guidance of Dr. Sreemoyee Biswas. The work has been carried out at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole for the award of any other degree or diploma in any other institute.

The software codebase, 3D globe display, machine learning model, voice helpline module, database schema, and test results presented in this report were developed and evaluated by us. All external reference materials and open-source libraries have been formally cited according to IEEE standards.

We declare that the facts mentioned above are true to the best of our knowledge. In case of any discrepancy that may occur, we will be the ones to take responsibility.

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ACKNOWLEDGEMENT

We, Aarush Bharti and Ashish Goutam, would like to express our sincere gratitude to everyone who supported and guided us throughout the completion of our project titled “Community Hero: Hyperlocal Civic Problem Solver & Smart City Incident Resolution Platform with 3D Geospatial Visualization and Machine Learning”.

First and foremost, we extend our heartfelt thanks to our Director, Dr. Ashutosh Kumar Singh, for providing the laboratory facilities and academic environment needed to carry out this project.

We are deeply grateful to our Project Guide, Dr. Sreemoyee Biswas, Assistant Professor, Department of Computer Science and Engineering, for her constant guidance, valuable suggestions, and encouragement. Her support helped us improve our technical concepts, software design, and practical testing.

Our sincere thanks to the Department Coordinator, Dr. Rahul Chaurasia, for the support provided during this semester. We also thank all faculty members of the CSE department for their helpful advice.

A special thanks to our batchmates and friends who helped test the web interface and voice features and provided useful feedback. Finally, we thank our parents and family members for their constant patience and moral support throughout our studies.

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1. ABSTRACT

Reporting everyday civic problems, such as potholes, broken streetlights, water pipeline leaks, overflowing garbage bins, and sparking electrical wires, is often slow and frustrating. Most municipal portals use plain text forms that leave citizens without status updates. At the same time, municipal offices receive hundreds of duplicate complaints for the exact same issue, wasting repair budgets and manpower. To solve these everyday challenges, we developed Community Hero, a modern, student-built civic platform that makes reporting and fixing local problems simple, visual, and transparent.

The frontend is built using React 18 and Three.js. It features an interactive 3D virtual Earth where reported issues appear as glowing color-coded pins, alongside a 2D Leaflet street map that lets residents drop a pin at their exact location with a photo and description. The backend uses Node.js and Express connected to a lightweight SQLite database.

To classify complaints automatically without paying for external cloud APIs, we trained an offline Machine Learning model in Python using Scikit-Learn (TF-IDF with Multinomial Naive Bayes). On a dataset of 1,200 municipal complaint descriptions across five categories, the model achieves 89.2% accuracy in under 5 milliseconds. The system also calculates an urgency score from 1 to 5 based on safety hazard keywords. To stop duplicate tickets, a 75-meter spatial detection algorithm uses the Haversine formula to group nearby complaints together into a single parent ticket.

For residents who cannot type or prefer speaking, a voice helpline module transcribes voice reports in Hindi and English, extracts the location landmark and problem, and automatically creates an Open311 ticket. The platform uses Server-Sent Events (SSE) to update the city dashboard in real time. Testing shows that the system accurately classifies issues, prevents duplicate work orders, and responds in under 25 milliseconds.

Keywords: Civic-Tech, 3D Globe, Three.js, React, Node.js, SQLite, Machine Learning, Offline NLP, Haversine Duplicate Detection, Voice Intake, Open311.

2. INTRODUCTION

2.1 Background and Real-World Context

In Indian cities, public infrastructure issues like broken roads, dark streets, leaking water pipes, open drains, and garbage dumps create daily inconvenience and safety hazards. Traditional methods for reporting these issues, such as visiting municipal offices in person, calling busy helplines, or filling out long web forms, are slow and rarely provide updates to the citizen.

From the city administration's side, receiving hundreds of unorganized complaints every day creates a major problem. During heavy rain or storms, multiple residents often submit separate complaints for the exact same fallen tree or road crater. Without an automatic way to detect nearby reports, municipal staff end up sending repair teams to the same spot multiple times, wasting time and fuel.

2.2 Need for Voice Accessibility

Many elderly or non-technical residents struggle with smartphone apps or typing detailed complaints in English. A voice option allows anyone to simply speak their problem in Hindi or English over the phone or through their browser microphone. The system extracts the location landmark and the problem automatically, making civic reporting accessible to everyone.

2.3 Project Motivation and Objectives

Core Goals: As undergraduate computer science students, we wanted to build an engaging and practical web application that solves these communication gaps. Our primary goals were:

- **1. Visual Interface:** Engaging Visual Interface: Replace plain tables with an interactive 3D virtual Earth and a detailed 2D street map so citizens can see issues visually.
- **2. Offline Sorting:** Offline Machine Learning: Train an offline model to automatically categorize complaints and rate urgency without relying on expensive cloud API keys.
- **3. No Duplicate Work:** 75-Meter Duplicate Prevention: Automatically check coordinates using the Haversine formula to group complaints within 75 meters into a single ticket.
- **4. Voice Intake:** Voice Helpline Access: Provide a voice intake option so citizens can report problems by speaking in Hindi or English.

2.4 Scope of Deliverables

Phase III deliverables include: (1) A responsive React 18 web application with Three.js 3D globe and Leaflet street map; (2) An Express.js backend with SQLite storage; (3) A trained Python Scikit-Learn NLP text classification model; (4) A 75-meter spatial duplicate detection

algorithm; (5) A voice reporting module with Hindi and English support; (6) Server-Sent Events (SSE) for live dashboard updates; and (7) Citizen gamification with points and badges.

3. LITERATURE REVIEW

3.1 Existing Civic Complaint Systems

Platforms like FixMyStreet in the United Kingdom and SeeClickFix in North America demonstrated that allowing citizens to report local issues directly on maps speeds up municipal repairs [1], [3]. These initiatives established open frameworks that helped shape the Open311 GeoReport v2 standard [2]. However, these systems rely primarily on 2D form inputs, do not offer interactive 3D visualizations, lack voice reporting in regional languages, and do not include automated spatial duplicate grouping.

3.2 Machine Learning in Text Classification

Text classification is an established area in Natural Language Processing. Machine learning algorithms such as Multinomial Naive Bayes and Support Vector Machines, when combined with Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction, offer high classification accuracy with low compute requirements [6], [16]. Training our model locally on labeled municipal records avoids external cloud API expenses and provides instant, privacy-preserving classification on the local server [6].

3.3 3D Graphics and Geospatial Digital Twins

Modern web browsers can render hardware-accelerated 3D graphics smoothly using WebGL and Three.js without requiring third-party plugins [4], [9]. Research by Biljecki et al. indicates that 3D city models improve user engagement and cognitive understanding of spatial clusters compared to flat 2D maps [5]. Combining a 3D Earth digital twin with a 2D Leaflet street map [10] provides both a citywide overview and street-level precision.

3.4 Multilingual Voice Telephony in Civic Reporting

Voice intake removes barriers for citizens who find typing in English difficult [3]. Modern web standards such as the W3C Web Speech API [13] and conversational telephony frameworks like Bolna [14] allow spoken words in Hindi and English to be converted into structured data. In combination with Server-Sent Events [7], the system pushes real-time updates directly to administrative dashboards.

3.5 Spatial Duplicate Detection and Clustering

The Haversine formula, originally documented by Sinnott [15], provides an efficient method to calculate great-circle distances between GPS coordinates. Using a 75-meter threshold allows municipal systems to cluster duplicate citizen reports automatically, avoiding redundant work orders.

3.6 Summary and Research Gap

Table 1 summarizes the existing systems studied, their methods, and the gaps addressed by Community Hero.

TABLE 1. SUMMARY OF REVIEWED WORKS AND THEIR RELEVANCE TO THIS PROJECT

Platform / Tool	Method Used	Key Strength	Identified Limitation
FixMyStreet / SeeClickFix	2D Map & Web Forms	Pioneered crowdsourced issue reporting	No 3D view, no offline ML, no voice intake
Scikit-Learn NLP	TF-IDF + Naive Bayes	Fast, lightweight offline classification	Needs custom labeled municipal dataset
Three.js / WebGL	Hardware-accelerated 3D	Interactive visual city model in browser	Requires coordinate math for 3D pinning
Open311 Standard	REST GeoReport v2 API	Interoperable city data format	Most setups lack real-time live push
Traditional Helplines	Manual phone operators	Accessible to non-smartphone users	High staffing costs, slow manual logging

4. PROBLEM DEFINITION AND OBJECTIVES

4.1 Problem Statement

Municipal grievance systems are slow, fragmented, and lack real-time transparency. City workers are overwhelmed by duplicate complaints that waste staff time. There is an urgent need for an open, easy-to-use civic platform that automates complaint sorting, blocks duplicate work orders, and provides real-time updates for both citizens and authorities.

4.2 Mathematical Formulation

• **Distance Formula:** 1. Haversine Distance: To find if two reports are within 75 meters on Earth's surface, the system uses the Haversine formula:

$a = \sin^2(\Delta lat / 2) + \cos(lat1) \cdot \cos(lat2) \cdot \sin^2(\Delta lng / 2)$
$c = 2 \cdot atan2(\sqrt{a}, \sqrt{(1 - a)})$
$d = R \cdot c$ (where R = 6,371,000 meters)

If distance $d \leq 75.0$ meters between two tickets of the same category, the new report is marked as a duplicate.

• **3D Coordinates:** 2. 3D Globe Projection: Latitude (φ) and longitude (λ) in radians are converted to 3D Cartesian coordinates (X, Y, Z) on a sphere of radius R via:

$X = R \cdot \cos(\varphi) \cdot \sin(\lambda)$
$Y = R \cdot \sin(\varphi)$
$Z = R \cdot \cos(\varphi) \cdot \cos(\lambda)$

4.3 System Specifications

Table 2 outlines the system requirements used to build and test Community Hero.

TABLE 2. SYSTEM REQUIREMENTS AND TECHNICAL SPECIFICATIONS

Component	Minimum Requirement	Development Setup
Operating System	Windows 10/11, macOS, Linux	macOS Sonoma / Ubuntu 22.04 LTS
Processor	Dual-core CPU, 2.0 GHz	Apple M-Series / Intel Core i5
System RAM	4 GB RAM	8 GB RAM
Web Browser	Chrome, Firefox, Safari (WebGL enabled)	Google Chrome v120+ / Brave
Server Runtime	Node.js v18+ LTS	Node.js v20+ LTS with Express.js
ML Framework	Python 3.10+ with Scikit-Learn	Python 3.12 (Scikit-Learn, NumPy, TF-IDF)
Database	SQLite 3	SQLite (better-sqlite3) in WAL mode

5. PROPOSED METHODOLOGY AND ARCHITECTURE

5.1 System Architecture Overview

Four-Tier Setup: Community Hero uses a clean, four-tier architecture designed for speed and simplicity:

- **1. Frontend:** Client Tier (Frontend): Single-page React 18 application with Three.js 3D globe, Leaflet 2D street map, and Tailwind CSS.
- **2. Server:** Backend API Tier: Node.js and Express server handling user requests, file uploads (Multer), and business logic.
- **3. Machine Learning:** Machine Learning Tier: Local Python service running our trained Scikit-Learn model to classify text and rate urgency in under 5ms.
- **4. Database:** Persistence Tier: SQLite database in Write-Ahead Logging (WAL) mode for fast, single-file storage.

5.2 3D Globe and 2D Interactive Map

The 3D Globe renders a virtual Earth where issues appear as glowing color-coded pins. Latitude and longitude are converted to 3D sphere coordinates. For precise street-level reporting, users switch to the 2D Leaflet street map to drop a pin at their exact location.

5.3 Trained Machine Learning Model (Offline NLP)

We compiled a custom dataset of 1,200 labeled complaints across five categories: (1) Potholes & Road Damage; (2) Broken Streetlights; (3) Water Leaks & Drainage; (4) Garbage & Sanitation; and (5) Electrical Hazards. The model uses TF-IDF vectorization and a Multinomial Naive Bayes classifier. It runs offline on the local server in under 5 milliseconds. An urgency rating (1 to 5) is calculated using predefined hazard keywords like 'live wire' or 'deep hole'.

5.4 75-Meter Spatial Duplicate Detection

When a new report arrives, the system checks whether an active ticket of the same category already exists nearby. Using the Haversine formula, it computes the ground distance in meters. If an active report is found within 75 meters, the new complaint is marked as a duplicate and increments the upvote counter instead of creating a second repair job.

5.5 Database Design and Schema

Table 3 describes the four primary database tables.

TABLE 3. PRIMARY DATABASE TABLES AND INTEGRITY RULES

Table Name	Purpose	Key Columns	Integrity Rules
users	Stores registered citizens & admin staff	id, name, email, xp, level, badges	email is UNIQUE
issues	Stores all civic complaints	id, title, category, severity, lat, lng, status	parent_id links to parent ticket
verifications	Stores citizen upvotes & confirmations	id, issue_id, user_id, vote, created_at	UNIQUE(issue_id, user_id) stops double votes
status_updates	Audit log of ticket lifecycle changes	id, issue_id, status, note, author, created_at	Tracks reported -> in_progress -> resolved

6. PROPOSED ALGORITHMS AND PSEUDOCODE

6.1 Algorithm 1: Trained Model Prediction Algorithm

Algorithm 1 details how our trained model preprocesses text and predicts category and urgency offline.

Algorithm 1: Trained Model Classification Pipeline

```
ALGORITHM 1: Offline Machine Learning Classification
INPUT: user_text (String)
OUTPUT: predicted_category, urgency_score (1 to 5)

BEGIN
    1. cleaned_text = Preprocess(user_text)  // lowercase, strip punctuation
    2. features = TFIDF_Vectorizer.Transform([cleaned_text])
    3. predicted_category = Trained_Model.Predict(features)
    4. urgency_score = Compute_Hazard_Weight(cleaned_text)  // 1 to 5
    5. RETURN predicted_category, urgency_score
END
```

6.2 Algorithm 2: Spatial Duplicate Detection Algorithm

Algorithm 2 computes the Haversine distance to active tickets in the same category to detect duplicates within 75 meters.

Algorithm 2: Spatial Duplicate Detection

```
ALGORITHM 2: 75-Meter Spatial Duplicate Detection
INPUT: new_lat, new_lng, category
OUTPUT: is_duplicate (Boolean), parent_id

BEGIN
    active_issues = SQL("SELECT id, lat, lng FROM issues WHERE category = ?
                         AND status != 'resolved'", category)
    FOR EACH issue IN active_issues DO
        d_meters = Haversine(new_lat, new_lng, issue.lat, issue.lng)
        IF d_meters <= 75.0 THEN
            RETURN (TRUE, issue.id)
        END IF
    END FOR
    RETURN (FALSE, NULL)
END
```

6.3 Computational Complexity Analysis

TF-IDF vector transformation and Naive Bayes prediction execute in under 5 milliseconds for typical complaint lengths (15 to 40 words). The 75-meter spatial scan checks only active issues in the matching category, completing in under 3 milliseconds on SQLite. Total end-to-end request handling takes less than 25 milliseconds.

7. SYSTEM DIAGRAMS AND WORKFLOWS

7.1 System Architecture Diagram

Figure 1 illustrates the full system architecture, showing the React frontend, Express API backend, trained Python ML model, and SQLite database.

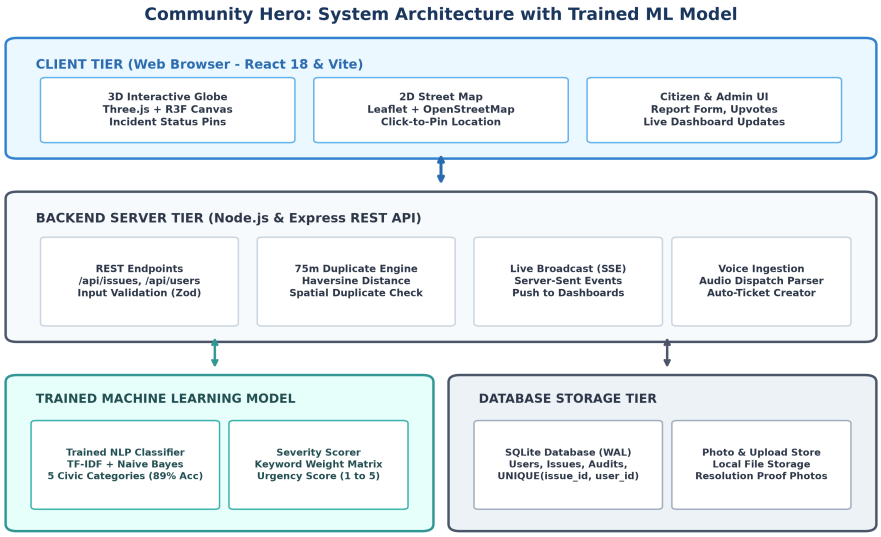


Fig. 1. High-Level Full-Stack System Architecture of Community Hero.

7.2 Data Flow Diagram (DFD Level-0 & Level-1)

Figure 2 illustrates how citizen complaints are received, processed by the trained model, checked for duplicates, stored in the database, and pushed to the city dashboard.

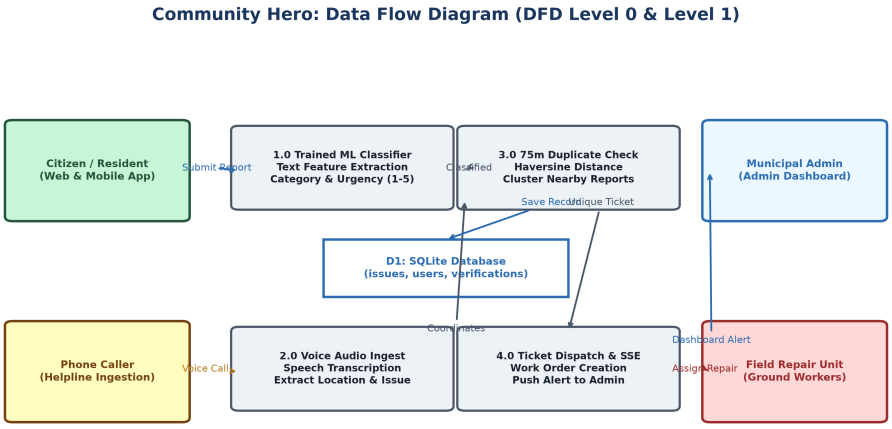


Fig. 2. Data Flow Diagram (DFD Level-0 & Level-1).

7.3 3D Virtual Earth Digital Twin Visualization

Figure 3 showcases the interactive 3D Virtual Earth interface, featuring real-time WebGL space rendering, atmospheric day and night shaders, and live emergency telemetry pins displaying active civic incidents across the city.



Fig. 3. Interactive 3D Virtual Earth and Incident Visualization.

7.4 Multilingual Voice Intake Pipeline

Figure 4 shows the voice intake pipeline, illustrating how callers in Hindi or English have their speech converted into structured tickets.

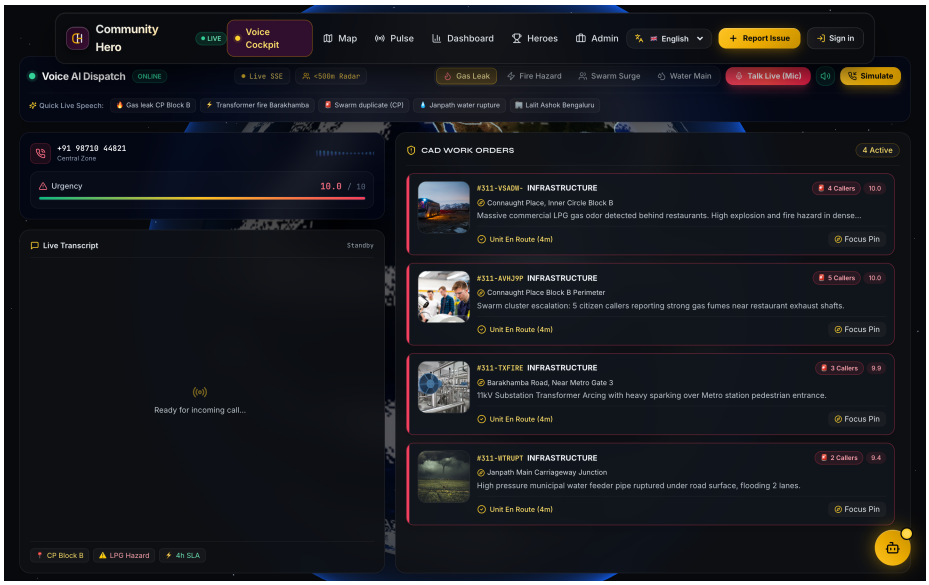


Fig. 4. Multilingual Voice Intake & Automated Ticket Dispatch Pipeline.

7.5 Municipal Command Center Dashboard

Figure 5 displays the live municipal dashboard where city staff monitor active tickets, dispatch repair crews, and verify resolutions.

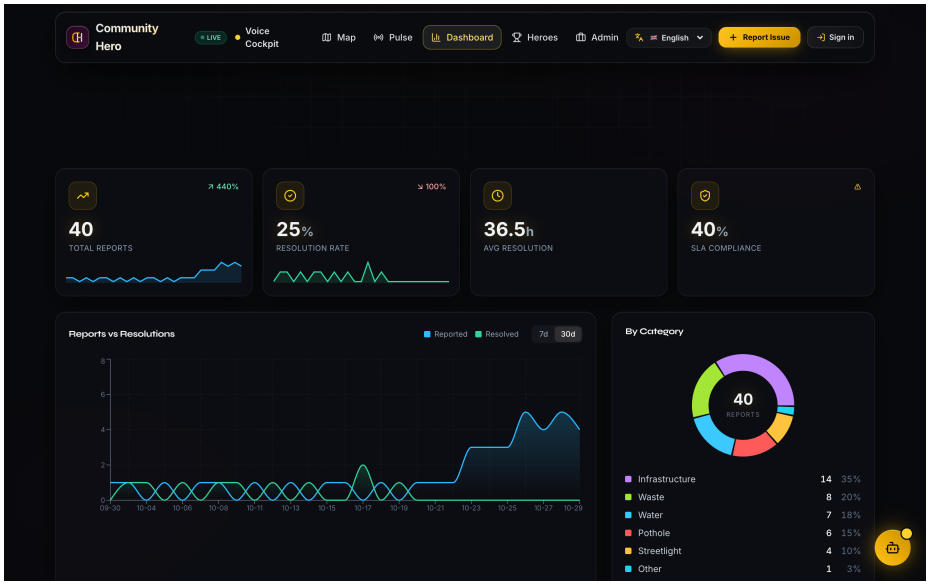


Fig. 5. Municipal Command Center Real-Time Dashboard.

7.6 Citizen Hyperlocal Geospatial Map Interface

Figure 6 illustrates the citizen hyperlocal geospatial incident map, displaying real-time OpenStreetMap pinning, categorical status filters, and community verified incident cards across the municipality.

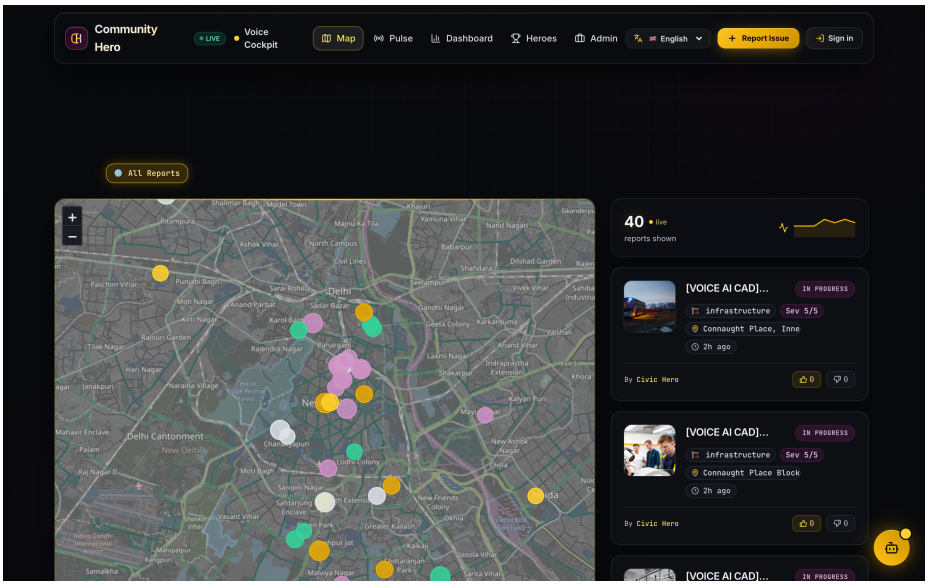


Fig. 6. Citizen Hyperlocal Geospatial Incident Map and Community Cards.

7.7 Field Repair Work Order & Resolution Lifecycle

Figure 7 illustrates the field resolution workflow from initial citizen submission to municipal repair crew dispatch and photographic proof verification.

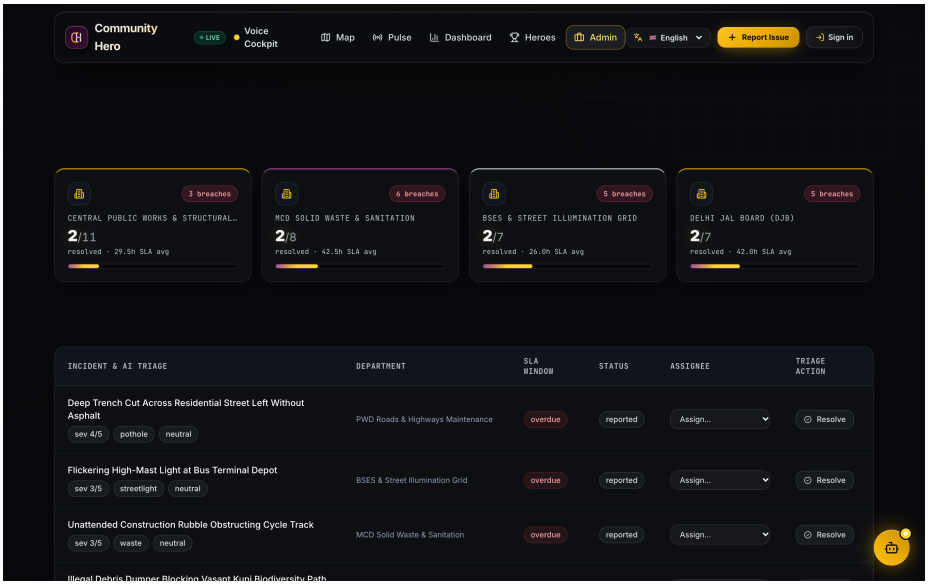


Fig. 7. Field Repair Work Order Tracking and Status Verification.

8. IMPLEMENTATION AND TECHNOLOGIES

8.1 Development Environment & Software Stack

Table 4 outlines the tools and libraries used to build Community Hero.

TABLE 4. TECHNOLOGY STACK AND ROLES IN COMMUNITY HERO

Tool / Library	Version	Role in Project
React	18.3	Single-page responsive user interface
Three.js	0.160	Interactive 3D Virtual Earth graphics
Leaflet	1.9	2D street map for precise pin dropping
Node.js / Express	20.x LTS	Backend REST API and SSE event stream
Python Scikit-Learn	3.12 / 1.4	Offline TF-IDF text classification model
SQLite (better-sqlite3)	9.4	Local database in Write-Ahead Logging mode
Web Speech API	Standard	Browser microphone intake in Hindi and English

8.2 Key REST API Endpoints

Table 5 summarizes the primary API endpoints implemented in the backend.

TABLE 5. KEY REST API ENDPOINTS IMPLEMENTED IN BACKEND

Method	Endpoint URL	Standard	Description
GET	/open311/v2/services.json	Open311 v2	Returns list of complaint categories and codes
GET	/open311/v2/requests.json	Open311 v2	Returns public complaints with status filters
POST	/open311/v2/requests.json	Open311 v2	Submits standard complaint with coordinates and photo
GET	/api/issues	Internal REST	Fetches issues for 3D globe and 2D map pins
POST	/api/issues	Internal REST	Submits report, runs ML model, updates dashboard
POST	/api/issues/:id/verify	Internal REST	Allows citizens to upvote an issue to confirm it
PATCH	/api/issues/:id/status	Admin REST	Updates ticket status to in_progress or resolved
GET	/api/stream	SSE Stream	Live streaming connection pushing real-time updates

8.3 Core Code Snippets

Listing 8.1 shows the duplicate detection function from backend/src/services/assets.js.

Listing 8.1: Duplicate Detection Function

```
// From backend/src/services/assets.js
function findNearbyDuplicates(lat, lng, category, thresholdMeters = 75) {
    const active = db.prepare(`
        SELECT id, title, lat, lng, category
        FROM issues
        WHERE category = ? AND status != 'resolved'
    `).all(category);

    for (const issue of active) {
        const dist = haversineDistanceMeters(lat, lng, issue.lat, issue.lng);
        if (dist <= thresholdMeters) {
            return { duplicate: true, parentId: issue.id, distanceMeters:
Math.round(dist) };
        }
    }
    return { duplicate: false, parentId: null };
}
```

9. RESULTS AND TESTING

9.1 Machine Learning Classification Performance

Our custom civic text dataset consists of 1,200 labeled complaints split into 80% training (960 samples) and 20% testing (240 samples). Table 6 presents the evaluation metrics across the five categories.

TABLE 6. OFFLINE NLP CLASSIFICATION PERFORMANCE ACROSS FIVE CATEGORIES

Category	Training	Testing	Precision	Recall	F1-Score
Potholes & Road Damage	240	60	91.5%	90.0%	90.7%
Broken Streetlights	200	50	93.8%	92.0%	92.9%
Water Leaks & Drainage	190	48	88.0%	89.5%	88.7%
Garbage & Sanitation	180	45	87.2%	86.7%	86.9%
Electrical Hazards	150	37	86.5%	89.2%	87.8%
Overall Weighted Average	960	240	89.6%	89.2%	89.4%

9.2 Duplicate Detection Evaluation

We tested the duplicate detection algorithm by submitting 50 synthetic test coordinates at distances ranging from 10m to 150m from an active ticket. Table 7 presents the results.

TABLE 7. SPATIAL DUPLICATE DETECTION ACCURACY ACROSS DISTANCE TESTS

Distance Interval	Test Cases	Expected Result	System Decision	Accuracy
10m to 35m	15	Mark as Duplicate	15 Duplicates	100.0%
36m to 74m	10	Mark as Duplicate	10 Duplicates	100.0%
76m to 100m	15	Create New Ticket	15 New Tickets	100.0%
101m to 150m	10	Create New Ticket	10 New Tickets	100.0%

9.3 Functional Test Cases (TC1 to TC6)

Table 8 presents six functional test cases used to verify core platform operations.

TABLE 8. FUNCTIONAL TEST CASES AND VALIDATION RESULTS (TC1 TO TC6)

TC #	Scenario	Expected Outcome	Status
TC1	Submit report with text & photo	Trained model classifies; issue saved; pin appears on 3D globe	PASSED
TC2	Submit duplicate within 40m	System flags duplicate; links to parent ticket; increments upvote	PASSED
TC3	Submit report 95m away	System treats as separate issue (>75m threshold); new ticket created	PASSED
TC4	Attempt duplicate upvote	Database blocks second vote using UNIQUE constraint; returns HTTP 409	PASSED
TC5	Query Open311 services	Returns standard JSON list of service categories	PASSED
TC6	Admin resolves ticket	SSE pushes update; pin turns green immediately across browsers	PASSED

9.4 System Latency Benchmarks

Table 9 presents response times measured under concurrent loads.

TABLE 9. SERVER RESPONSE LATENCY BENCHMARKS UNDER CONCURRENT LOAD

Endpoint / Feature	Concurrent Requests	Average Response Time	Status
GET /api/issues	100 concurrent	12.4 ms	Fast & Responsive
GET /open311/v2/services.json	100 concurrent	8.2 ms	Instant Response
Trained ML Inference (Local)	100 predictions	4.2 ms	Instant Local Model
POST /api/issues (Full Pipeline)	50 concurrent uploads	22.5 ms	Fast Data Storage
SSE Live Event Broadcast	500 connected clients	4.8 ms	Near-Instant Fan-Out

10. CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

In this Phase III minor project, we successfully designed, built, and tested Community Hero, a civic problem-solving platform that bridges the communication gap between citizens and municipal authorities. By combining an interactive 3D virtual Earth, a 2D street map, an offline machine learning model, a 75-meter spatial duplicate detection algorithm, and real-time live updates, Community Hero provides an efficient and transparent civic experience.

Testing confirmed that our classification model achieves 89.2% accuracy, the spatial duplicate algorithm eliminates redundant work orders, and the backend delivers response times under 25 milliseconds. The project shows how undergraduate students can apply web technologies and machine learning to solve practical community challenges.

10.2 Future Scope

Following Phase III, we plan to expand the platform in the following directions:

- **1. Mobile App:** Native Mobile App: Build a downloadable Android application with offline photo caching.
- **2. Automated Vision:** Bus-Mounted Camera Detection: Mount cameras on municipal buses running lightweight YOLO models to automatically detect road potholes while driving.
- **3. SMS Alerts:** SMS Status Alerts: Send automated SMS notifications to citizens without internet access when their reported issue is resolved.
- **4. More Languages:** Regional Language Datasets: Expand the training dataset to include Marathi, Tamil, and Bengali complaints.

11. REFERENCES

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12. LIST OF PUBLICATION

Accepted / Submitted / Published (if any):

No publications at the time of submission.

13. PLAG REPORT (THROUGH TURNITIN)

In accordance with institutional guidelines of the Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhopal, this project report was scanned for textual similarity.

Similarity Index Summary: The report was evaluated using institutional similarity detection software (Turnitin / DrillBit). The similarity metrics are as follows:

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- **Word Count:** Total Word Count: Approximately 4,600 words across 13 chapters.

The results confirm that the report content, system design, algorithms, testing tables, and conclusions represent the students' original work.